Code No: 157CE **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
INTRODUCTION TO EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Describe the characteristics of an embedded system in detail.
b) Explain about the top down and bottom up design approach of an embedded system.[7+8]

2.a) What is an Embedded system? List out the differences between an embedded system and a general purpose computer?
b) What is operational quality attribute? Explain the importance of non-operational quality attributes to be considered in any embedded system. [7+8]

3.a) What are the different types of memories used in embedded system? Explain each with an example.
b) Explain the role of actuator in embedded system design. [8+7]

4.a) Discuss different communication interfaces with respect to embedded systems?
b) What is the core of embedded system? Explain. [8+7]

5. What are the different languages used for embedded firmware development? Explain.[15]

6. What are the various design approaches of embedded firmware? [15]

7.a) What is a process? With a neat representation explain the process states and state transition?
b) What is a Task and briefly explain about the task scheduling mechanism? [8+7]

8.a) What is debugging and explain about debugging with In Circuit Emulator.
b) Explain integration approach of hardware and firmware. [8+7]

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